pyvar – A Python Package for Market Risk Management*

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Abstract

This paper presents `pyvar`, a Python-based package for the estimation, visualization, and backtesting of Value-at-Risk (VaR) and Expected Shortfall (ES) in the context of market risk management. The package implements a comprehensive suite of models covering nearly all academic and industry-standard approaches, including parametric, non-parametric, and simulation-based methods. Both univariate and multivariate portfolio frameworks are supported, alongside advanced capabilities such as GARCH models with non-normal innovations, Extreme Value Theory (EVT) tail modeling, filtered and weighted historical simulations, and time-varying correlation estimation using principal component analysis and Ledoit–Wolf shrinkage. Using equity and options data sourced from Yahoo Finance, our empirical analysis shows that `pyvar` delivers benchmark-level performance consistent with industry standards. We implement the three standard backtesting procedures — the Kupiec test for unconditional coverage, the Christoffersen test for independence, and the joint test — and find that many of our models are able to consistently pass all three across different portfolios and market conditions. By combining accessibility, transparency, and reproducibility, `pyvar` offers a robust computational framework suitable for both academic research and professional risk management practice.

Keywords: Value-at-Risk (VaR), Expected Shortfall (ES), Monte Carlo Simulation, GARCH Model.

1 Introduction

The measurement and management of market risk — the risk of losses on trading positions due to adverse market movements — has been a central concern for regulators and practitioners for decades. Since the Basel Committee's 1996 Market Risk Amendment, Value-at-Risk (VaR) has served as the regulatory benchmark for capital requirements, typically defined as the 1% quantile of portfolio losses over a fixed horizon. In daily practice, VaR is computed at multiple confidence levels (95% or 99%) and horizons (1–10 days), providing a standard metric for risk control.

From a statistical perspective, VaR is the quantile of the conditional profit-and-loss (P&L) distribution. Estimating this quantile is challenging because financial returns exhibit volatility clustering, heavy tails, and skewness. Simple variance–covariance approaches under Gaussian assumptions therefore tend to underestimate risk, motivating the use of more robust methods such as historical simulation, filtered historical simulation (FHS), Extreme Value Theory (EVT), and GARCH-type volatility models with non-normal innovations.

Over the past three decades, research and practice have converged on a broad toolkit for VaR and Expected Shortfall (ES) estimation, incorporating time-varying correlations, factor structures, and simulation-based methods. These developments have enhanced the accuracy of risk measurement for both linear portfolios and portfolios containing derivatives.

In this context, we introduce `pyvar`, a Python package designed for market risk management with a focus on equity and options portfolios. The package automates the entire pipeline — portfolio construction, model estimation, risk metric computation, visualization, and backtesting — and implements parametric, non-parametric, and simulation-based approaches. It supports advanced features such as GARCH models with non-normal innovations, EVT tail modeling, weighted historical simulations, and dynamic correlation estimation using principal components and Ledoit–Wolf shrinkage.

Validation is central: `pyvar` includes the three standard backtesting procedures — the Kupiec test for unconditional coverage, the Christoffersen independence test, and their joint test — and many models consistently pass all three across portfolios and market conditions. This paper outlines the theoretical background, implementation details, and empirical performance of the models, with replication code available on GitHub: `https://github.com/Alessandro-Dodon/pyvar`.

2 Basic VaR Models

We begin with standard approaches to computing VaR and Expected Shortfall (ES). Let $c \in (0, 1)$ denote the confidence level, and define $\alpha = 1 - c$ as the corresponding probability of a large loss. Value-at-Risk (VaR) at level α is defined as the worst loss L over a fixed horizon such that the probability of observing a larger loss is at most α:

$$\Pr(L > \mathrm{VaR}_\alpha) \leq \alpha.$$

Intuitively, this means we are confident that with probability c, losses will not exceed VaR_α over the given horizon.

By convention, VaR is expressed as a positive monetary amount. To simplify the notation—especially in univariate models—we omit the scaling by invested wealth W and use percentage-based notation instead. In the package, wealth is considered optional in univariate models, so percentage-based VaR is the default output. If specified, monetary VaR can be computed easily. In portfolio-level models, wealth is generally inferred from the parameters, and VaR is reported both in monetary and percentage terms. The only exception is simulation-based models, where VaR is returned exclusively in monetary units. The same notation conventions apply in exactly the same way to ES.

To simplify the calculations we assume that W is strictly positive and that returns follow a zero-mean process. We define the loss variable as $\ell_t = -r_t$. We start with models that assume constant volatility over time.

2.1 Normal, Student-t and Cornish-Fisher VaR

The parametric normal and Student-t VaR are computed as:

$$\mathrm{VaR}_\alpha^{\text{normal}} = z_\alpha \sigma, \qquad \mathrm{VaR}_\alpha^{t_\nu} = q_{t_\nu,\alpha}\sigma,$$

where $z_\alpha = \Phi^{-1}(1-\alpha)$ is the upper-tail quantile of the standard normal distribution and $q_{t_\nu,\alpha} = F_{t_\nu}^{-1}(1-\alpha)$ is the upper-tail quantile of the Student-t distribution with ν degrees of freedom. Parameters are estimated via maximum likelihood.

Normal VaR is rarely reliable for daily financial data, as it tends to underestimate extreme losses. In our empirical experiments, Student-t models offer significantly better tail fit and risk estimation.

Expected Shortfall (ES), also known as Conditional VaR, for these models is computed as:

$$\mathrm{ES}_\alpha^{\text{normal}} = \sigma \cdot \frac{\phi(z_\alpha)}{\alpha}, \qquad \mathrm{ES}_\alpha^{t_\nu} = \sigma \cdot \frac{f_{t_\nu}(z_\alpha)}{\alpha} \cdot \frac{\nu + z_\alpha^2}{\nu - 1},$$

where ϕ is the standard normal density and f_{t_ν} is the density of the Student-t distribution.

The Cornish-Fisher expansion adjusts the normal quantile to account for higher-order moments of the return distribution — particularly skewness and kurtosis. This leads to:

$$\mathrm{VaR}_\alpha^{\text{CF}} = -z_\alpha^{\text{CF}} \cdot \sigma,$$

where:

$$z_\alpha^{\text{CF}} = z_\alpha + \tfrac{1}{6}(z_\alpha^2 - 1)s + \tfrac{1}{24}(z_\alpha^3 - 3z_\alpha)k - \tfrac{1}{36}(2z_\alpha^3 - 5z_\alpha)s^2,$$

with s and k representing the sample skewness and excess kurtosis, and z_α the standard normal upper-tail quantile. This expansion allows for asymmetry and fat tails in a parametric way, though it does not extend naturally to a consistent ES estimate. It is only effective when deviations from normality are mild.

2.2 Historical and Hybrid VaR

The non-parametric (historical) VaR is computed as:

$$\mathrm{VaR}_\alpha^{\text{hist}} = Q_\alpha(\ell_t),$$

where $Q_\alpha(\ell_t)$ is the empirical α-quantile of losses.

Expected Shortfall for the historical method is given by:

$$\mathrm{ES}_\alpha^{\text{historical}} = \mathbb{E}\left[\ell_t \mid \ell_t \geq \mathrm{VaR}_\alpha\right].$$

The Hybrid or Exponentially Weighted Historical approach refines this method by applying exponentially decaying weights:

$$w_t = \alpha_0 \cdot \lambda^t, \quad \text{where } \alpha_0 = \frac{1-\lambda}{\lambda(1-\lambda^n)},$$

for $t = 0, 1, \ldots, n-1$, and $\lambda \in (0,1)$ is the decay factor.

Weighted VaR and ES are computed as:

$$\mathrm{VaR}_\alpha^{\mathrm{hybrid}} = Q_\alpha^{\mathrm{w}}(\ell_t), \qquad \mathrm{ES}_\alpha^{\mathrm{hybrid}} = \frac{\sum_{t:\,\ell_t \geq \mathrm{VaR}_\alpha} w_t \ell_t}{\sum_{t:\,\ell_t \geq \mathrm{VaR}_\alpha} w_t}.$$

This weighting scheme improves responsiveness to changing volatility conditions and mitigates the ghost effect — where outdated extreme events inflate risk estimates long after they are relevant.

3 Extreme Value Theory

Technically still a parametric model, EVT is more sophisticated than the methods discussed above. We implement it with the Peaks-Over-Threshold (POT) approach: once a high threshold u (e.g., the 99th-percentile loss) is chosen, a Generalized Pareto Distribution (GPD) is fitted to the excesses above u.

VaR and ES are then derived analytically from the fitted GPD parameters. This allows for more accurate estimation of rare, extreme losses. The formulas used are:

$$\mathrm{VaR}_\alpha = u + \frac{\beta}{\xi}\left[\left(\tfrac{n}{n_u}(1-c)\right)^{-\xi} - 1\right], \qquad \mathrm{ES}_\alpha \;\; = \frac{\mathrm{VaR}_\alpha + \beta - \xi u}{1-\xi},$$

where ξ and β are the estimated shape and scale parameters, n is the sample size, and n_u is the number of exceedances.

However, the accuracy of this approach depends heavily on having a sufficiently large dataset. Fitting the tail of the distribution requires enough exceedances over the threshold to reliably estimate the shape and scale parameters. When used appropriately, EVT enables estimation of VaR and ES at extremely high confidence levels (e.g., 99.97% or 99.99%), which are nearly impossible to obtain with standard parametric or nonparametric methods.

In one of our example notebooks, we demonstrate how this method becomes particularly powerful when combined with simulations (e.g., Monte Carlo). Profit and loss scenarios can be generated using any of our simulation methods, which will be described in detail below. This setup ensures an abundance of tail data, allowing EVT to fit the tail parameters more effectively and yield robust estimates of extreme risk measures.

4 Volatility Models

Volatility models capture time-varying risk through conditional variance dynamics. In all cases, we use a semi-empirical approach for VaR computation:

$$\mathrm{VaR}_t = -\sigma_t\, z_\alpha, \qquad z_t = \frac{r_t}{\sigma_t},$$

where σ_t is the volatility estimate from the chosen model and z_α is the lower-tail α-quantile from the empirical distribution of standardized innovations z_t. This framework preserves the model-based conditional variance while allowing tail behavior to be estimated directly from the data.

We group volatility models into two categories: simple rolling estimators, which are fast and easy to implement, and ARCH/GARCH-type models, which provide a parametric structure for modeling and forecasting volatility. Realistic conditional volatility modeling is essential: without accounting for time-varying volatility, VaR estimates systematically fail the independence test — a critical requirement for model validity. This makes volatility models a core component of modern risk estimation pipelines.

4.1 Rolling Estimators

Rolling estimators use past return data to compute volatility over a moving window or with exponentially decaying weights. They serve as benchmarks due to their simplicity and computational efficiency:

$$\sigma_t^{2,\mathrm{MA}} = \frac{1}{n}\sum_{i=1}^{n} r_{t-i}^2, \qquad \sigma_t^{2,\mathrm{EWMA}} = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2.$$

The moving average (MA) estimator computes the unweighted average of squared returns over a fixed window of n periods, while the exponentially weighted moving average (EWMA) assigns greater weight to more recent returns through the decay factor λ. Both can be combined directly with empirical quantiles of z_t to produce simple, volatility-adjusted VaR and ES measures.

4.2 ARCH and GARCH-Type Models

ARCH-type models specify conditional variance as a function of past squared returns. The ARCH(p) model is given by:

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \, r_{t-i}^2.$$

The GARCH(1,1) model extends this by adding a lagged variance term:

$$\sigma_t^2 = \omega + a\, r_{t-1}^2 + b\,\sigma_{t-1}^2.$$

In both cases, returns are modeled as:

$$r_t = \sigma_t z_t,$$

where z_t are i.i.d. standardized innovations. For simplicity, we set the mean return to zero ($\mu = 0$), consistent with our earlier assumptions.

Multi-step forecasts for the GARCH(1,1) model are available in closed form:

$$\begin{aligned}
\mathbb{E}[\sigma_{t+\tau}^2] &= \mathrm{VL} + (a+b)^\tau(\sigma_t^2 - \mathrm{VL}),\\
\mathbb{E}[\sigma_{t,T}^2] &= \mathrm{VL}\left(T - 1 - \frac{(a+b)(1-(a+b)^{T-1})}{1-(a+b)}\right) + \sigma_t^2\,\frac{1-(a+b)^T}{1-(a+b)},
\end{aligned}$$

where $\mathrm{VL} = \frac{\omega}{1-a-b}$ is the long-run variance. In our framework, if desired, VaR is easily obtained by scaling the forecasted conditional variance with the empirical lower-tail quantile of standardized innovations z_t, consistent with our earlier definition.

Several extensions are supported. The EGARCH(1,1) model captures leverage effects and ensures positivity of variance by modeling the log variance:

$$\log(\sigma_t^2) = \omega + \beta\log(\sigma_{t-1}^2) + \alpha\frac{|r_{t-1}|}{\sigma_{t-1}} + \gamma\frac{r_{t-1}}{\sigma_{t-1}}.$$

The GJR-GARCH(1,1) model introduces an asymmetry term for negative shocks:

$$\sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \gamma\, I_{\{r_{t-1}<0\}}\, r_{t-1}^2 + \beta\,\sigma_{t-1}^2,$$

where $I_{\{.\}}$ is the indicator function. The APARCH(1,1) model allows for a flexible power δ and an asymmetry parameter γ:

$$\sigma_t^\delta = \omega + \alpha\left(|r_{t-1}| - \gamma r_{t-1}\right)^\delta + \beta\sigma_{t-1}^\delta.$$

All our GARCH-type models can be estimated via maximum likelihood under Normal, Student-t_ν, GED, or Skew-t_ν innovations.

4.3 Expected Shortfall in Volatility Models

For all volatility models, ES is computed as:

$$\mathrm{ES}_t = -\sigma_t \, \mathbb{E}[z_t \mid z_t < z_\alpha],$$

where the expectation is taken with respect to the empirical distribution of standardized innovations z_t. This semi-empirical approach allows for flexible tail estimation while preserving the conditional variance structure.

The importance of realistic volatility modeling becomes especially clear in backtesting: without accounting for time-varying volatility, VaR estimates often fail the independence test, a key requirement for model validity. This makes volatility models a core component of modern market risk estimation pipelines.

5 Time-Varying Correlation Models

Time-varying correlation models extend volatility modeling to the multivariate setting, enabling portfolio-level risk estimation. Portfolio volatility is computed directly from positions and the conditional covariance matrix as

$$\sigma_{p,t} = \sqrt{w_t^\top \Sigma_t w_t}, \qquad w_t = \frac{x_t}{\sum_{i=1}^N x_{i,t}},$$

where $x_t = (x_{1,t}, \ldots, x_{N,t})^\top$ is the vector of monetary positions (allowing for short positions), w_t the portfolio weights, and Σ_t the estimated conditional covariance matrix.

This conditional volatility serves as the scale factor for both Value-at-Risk and Expected Shortfall:

$$\mathrm{VaR}_t = -\sigma_{p,t} \, z_\alpha, \qquad \mathrm{ES}_t = -\sigma_{p,t} \, \mathbb{E}[z_t \mid z_t < z_\alpha],$$

where z_t are standardized innovations and z_α the α-quantile of their distribution.

Two estimation approaches are possible. In the parametric case, z_α is taken from the standard normal distribution; in this setting, the formulas are adjusted consistently with our earlier definitions in the Basic VaR section, where the upper-tail quantile of the normal distribution is used. In the semi-empirical case, z_α and the tail expectation are estimated directly from the empirical distribution of z_t. The semi-empirical approach requires additional stability conditions on portfolio weights: they must sum to one, not be too small in absolute value, and avoid excessive short positions.

This framework naturally integrates into all correlation-based estimators, assuming a buy-and-hold strategy for positions.

5.1 Rolling Estimators

The simplest approach is to compute Σ_t from past returns over a moving window or using exponentially decaying weights:

$$\Sigma_t^{\mathrm{MA}} = \frac{1}{n} \sum_{i=1}^{n} r_{t-i} r_{t-i}^\top, \qquad \Sigma_t^{\mathrm{EWMA}} = \lambda \, \Sigma_{t-1} + (1-\lambda) r_{t-1} r_{t-1}^\top,$$

where r_t is the vector of asset returns and $\lambda \in (0, 1)$ is the decay factor. These estimators are computationally fast and adapt to changing market conditions, but they can become unstable when the number of assets is large relative to the sample size.

5.2 Shrinkage and Dimensionality Reduction Methods

To address instability in high-dimensional covariance estimation, we implement two methods: rolling principal component analysis (PCA) and Ledoit–Wolf shrinkage.

In the PCA approach, we compute a rolling sample covariance matrix and approximate it by retaining only the top k eigencomponents:

$$\Sigma_t^{\mathrm{PCA}} \approx V_t \Lambda_t V_t^\top,$$

where V_t contains the leading eigenvectors and Λ_t the corresponding eigenvalues. This yields a low-rank, de-noised estimate of Σ_t.

Ledoit–Wolf shrinkage is likewise applied on a rolling window. At each step, the sample covariance matrix S_t is shrunk toward a structured target F:

$$\Sigma_t^{\mathrm{LW}} = \delta_t F + (1 - \delta_t) S_t,$$

where F is typically chosen as the identity scaled by the average variance, and $\delta_t \in [0, 1]$ is the shrinkage intensity estimated from the data. This method regularizes noisy covariance estimates while preserving key features of the covariance structure.

Both PCA and Ledoit–Wolf shrinkage are particularly effective for large portfolios, where the sample covariance matrix is often noisy or ill-conditioned. They improve the stability of VaR and ES estimates, especially when used in real-time risk monitoring applications.

6 Factor Models

Factor models are a widely used approach in finance for estimating portfolio-level VaR and ES. While fundamentally statistical, they are guided by financial theory, linking asset returns to a small number of systematic risk drivers with clear economic interpretations.

This method—often referred to as risk mapping—uses estimated factor exposures to represent the portfolio's sensitivity to broad market forces. We implement both the Sharpe single-factor model and the Fama-French three-factor model, capturing market, size, and value effects through estimated betas.

6.1 Sharpe Single-Factor Model

For the Sharpe Model, we assume that each asset's return r_i follows the linear factor structure:

$$r_i = \alpha_i + \beta_i r_m + \varepsilon_i,$$

where r_m is the return on the market portfolio, β_i is the asset's sensitivity to the market factor, α_i is the asset's abnormal return, and ε_i is a zero-mean idiosyncratic shock uncorrelated with r_m and other residuals. The market return r_m has variance σ_m^2, and $\mathrm{Var}(\varepsilon_i) = \sigma_{\varepsilon_i}^2$. We support both a simplified setting with constant market volatility and a dynamic version where σ_m^2 is estimated via a GARCH model. Portfolio weights are assumed to be perfectly constant.

The variance of an individual asset i is given by:

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{\varepsilon_i}^2.$$

The covariance matrix of asset returns can be written compactly as:

$$\Sigma = \beta\beta^\top \cdot \sigma_m^2 + D,$$

where β is the $N \times 1$ vector of factor loadings, and D is the diagonal matrix of idiosyncratic variances.

The total portfolio variance is then:

$$\sigma_p^2 = w^\top \Sigma w.$$

Finally, under the normality assumption, portfolio VaR and ES are given by:

$$\mathrm{VaR}_\alpha = z_\alpha \, \sigma_p, \qquad \mathrm{ES}_\alpha = \sigma_p \, \frac{\phi(z_\alpha)}{\alpha}.$$

6.2 Fama-French Three-Factor Model

The Fama-French three-factor model extends the Sharpe model by incorporating two additional systematic risk factors beyond the market: size (SMB, small minus big) and value (HML, high minus low book-to-market ratio). This framework captures the tendency for small-cap and value stocks to generate excess returns relative to the market, improving the explanatory power for cross-sectional differences in asset returns.

Each asset's excess return is modeled as:

$$r_i - r_f = \alpha_i + \beta_{i,1} \cdot \text{Mkt-RF} + \beta_{i,2} \cdot \text{SMB} + \beta_{i,3} \cdot \text{HML} + \varepsilon_i,$$

where r_f is the risk-free rate, and ε_i is the idiosyncratic error. We estimate factor loadings $\beta_{i,k}$ through OLS regression.

Let B be the $N \times 3$ matrix of estimated factor loadings where the i-th row is $(\beta_{i,1}, \beta_{i,2}, \beta_{i,3})$, Σ_f the 3×3 sample covariance matrix of the factors, and D the diagonal matrix of residual variances $\sigma_{\varepsilon_i}^2$. The total portfolio variance is:

$$\sigma_p^2 = w^\top \left(B \Sigma_f B^\top + D \right) w.$$

Once the portfolio variance is known, we compute VaR and ES as we did for the single factor model.

7 Analytic VaR

Although analytic VaR relies on strong assumptions—most notably that asset returns are normally distributed and follow an independent and identically distributed (i.i.d.) process—it remains a widely used and insightful tool in risk management. Its closed-form nature allows for fast computation and enables a detailed decomposition of risk, not only for measurement but also for understanding risk contributions and managing exposures.

In this section, we compute several analytic VaR measures using the previously defined monetary position vector $x_t = (x_{1,t}, \ldots, x_{N,t})^\top$ and the covariance matrix Σ.

The quantile z_α corresponds to the desired tail probability level and is always derived from the standard normal distribution, as the underlying assumption is that asset returns are normally distributed.

The asset-normal parametric portfolio VaR is defined as:

$$\mathrm{VaR}_t = z_\alpha \cdot \sqrt{x_t^\top \Sigma x_t} \cdot \sqrt{h}.$$

Here, h denotes the holding period, under the assumption of i.i.d. returns.

Assuming all asset returns are perfectly correlated ($\rho = 1$) and there are no short positions, the undiversified portfolio VaR is simply the sum of the individual asset VaRs:

$$\mathrm{UVaR}_t = \sum_{i=1}^{N} \mathrm{VaR}_{i,t}.$$

The marginal VaR, representing the sensitivity of portfolio VaR to a small change in position $x_{i,t}$, is given by:

$$\Delta\mathrm{VaR}_{i,t} = \mathrm{VaR}_t \cdot \frac{(\Sigma x_t)_i}{x_t^\top \Sigma x_t}.$$

It measures the change in total VaR from adding one extra unit of asset i.

The component VaR is:

$$\mathrm{CVaR}_{i,t} = x_{i,t} \cdot \Delta\mathrm{VaR}_{i,t},$$

which captures the absolute contribution of asset i to total portfolio VaR.

The relative component VaR is:

$$\mathrm{RCVaR}_{i,t} = \frac{\mathrm{CVaR}_{i,t}}{\mathrm{VaR}_t},$$

interpreted as the percentage share of total VaR attributable to asset i.

For a vector a representing changes in the portfolio allocation, the incremental VaR is:

$$\mathrm{IVaR}_t = \Delta\mathrm{VaR}_t^\top \cdot a,$$

which estimates the change in total VaR resulting from shifting the portfolio by a.

Each of these VaR metrics can be complemented with a corresponding Expected Shortfall (ES), computed under the normality assumption.

8 Options

We now turn to portfolios containing options. Delta-Normal approximations provide fast and analytically tractable methods for estimating the VaR of portfolios containing European options. These methods rely on the Black-Scholes model, which not only underpins the delta-based risk approximation used here but also serves as a foundation for the simulation-based techniques discussed in the next section.

This approach is another form of risk mapping: it translates the non-linear risk of options into equivalent linear exposures (deltas) to the underlying assets, allowing standard portfolio VaR techniques to be applied.

8.1 Black–Scholes Formula

The Black–Scholes model assumes that the underlying asset follows a geometric Brownian motion (GBM):

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t,$$

where μ is the drift, σ the volatility, and dW_t is a Wiener increment with $dW_t \sim \mathcal{N}(0, dt)$. Over a finite step Δt, the exact solution is:

$$S_{t+\Delta t} = S_t \exp\Big((\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon\Big), \qquad \varepsilon \sim \mathcal{N}(0,1).$$

which implies lognormal dynamics for S_t. Based on this assumption, the fair value of a European call or put option is given by:

$$C = N(d_1)S_t - N(d_2)Ke^{-r(T-t)}, \qquad P = N(-d_2)Ke^{-r(T-t)} - N(-d_1)S_t,$$

where:

$$d_1 = \frac{\ln(S_t/K) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}, \qquad d_2 = d_1 - \sigma\sqrt{T-t}.$$

Here, S_t is the spot price, K the strike price, $T-t$ the time to maturity in years, r the continuously compounded risk-free rate, σ the annualized volatility of returns, and $N(\cdot)$ the standard normal cumulative distribution.

8.2 Option Delta Method

The option's delta, which measures sensitivity to the underlying asset, is given by $\Delta = N(d_1)$ for a call and $\Delta = N(d_1) - 1$ for a put. Using this, a first-order approximation of the VaR_α for a single option position with q contracts, each on N units of the underlying asset, is:

$$\mathrm{VaR}_\alpha = z_\alpha \cdot |\Delta| \cdot S \cdot \sigma \cdot q \cdot N \cdot \sqrt{h},$$

where z_α is the upper-tail quantile of the standard normal distribution corresponding to the confidence level and h denotes the holding period as previously defined in the Analytic VaR section.

For a portfolio of options on multiple underlying assets, we compute the dollar delta exposure for each asset as $x_i = \Delta_i \cdot S_i \cdot q_i \cdot N_i$, and estimate VaR_α using the variance-covariance matrix Σ of asset returns:

$$\mathrm{VaR}_\alpha = z_\alpha \cdot \sqrt{\mathbf{x}^\top \Sigma \mathbf{x}} \cdot \sqrt{h}.$$

These formulas provide a fast approximation of market risk, assuming that portfolio values are linearly sensitive to underlying prices and that asset returns are normally distributed. However, when portfolios contain complex or highly nonlinear positions, these assumptions break down. In such cases, simulation-based approaches—which will be introduced next—offer a more flexible and accurate alternative.

9 Simulation Methods

Simulation methods, or full valuation approaches, are crucial when dealing with portfolios that carry substantial derivative exposure or exhibit pronounced non-linear behavior—such as those dominated by options. Unlike analytic methods that rely on linear approximations, these techniques reprice the full portfolio under a wide range of simulated scenarios, making them indispensable for capturing complex risk dynamics that traditional models cannot.

In our package, these methods enable risk estimation for portfolios that include both equity and options, as well as for equity-only or options-only portfolios. This flexibility is demonstrated in the notebook examples.

By default, all simulation methods estimate one-day ahead VaR, ES, and simulated profit and loss (P&L). Some multi-day extensions are also supported, enabling risk forecasting over longer horizons.

Our simulation methods use the same GBM framework in discrete time. We work with daily log-returns:

$$r_{t+1} \sim \mathcal{N}(\mu, \Sigma), \qquad S_{t+1} = S_t \exp(r_{t+1}),$$

where μ and Σ are directly estimated from empirical daily log-returns.

We begin with the one-day Monte Carlo approach under the assumption of multivariate normality. Future returns are simulated as:

$$\mu = \mathbb{E}[r_t], \quad \Sigma = \mathrm{Cov}(r_t), \quad L = \mathrm{Cholesky}(\Sigma),$$

$$r^{(i)} = \mu + Lz^{(i)}, \quad z^{(i)} \sim \mathcal{N}(0, I),$$

$$S^{(i)} = S_0 \circ \exp(r^{(i)}).$$

where $\circ$ denotes element-wise multiplication. Portfolio profit and loss is calculated as:

$$\Delta P^{(i)} = w^\top (S^{(i)} - S_0) + \sum_{j=1}^{N_{\mathrm{opt}}} q_j \left(C(S_j^{(i)}, \tau_j') - C_{j,0} \right),$$

where q_j is the number of option contracts, $C(\cdot)$ is the Black-Scholes option price, and $\tau_j' = \max(T_j - \frac{1}{252}, 0)$ is the adjusted time to maturity.

We define simulated portfolio losses as $\ell^{(i)} = -\Delta P^{(i)}$. From the empirical distribution $\{\ell^{(i)}\}_{i=1}^{N}$, where N is the number of simulations, we compute:

$$\mathrm{VaR}_\alpha = Q_\alpha(\ell), \qquad \mathrm{ES}_\alpha = \mathbb{E}[\ell \mid \ell \geq \mathrm{VaR}_\alpha].$$

Note that since the simulations are based on portfolio profit and loss, the resulting VaR and ES estimates are expressed directly in monetary terms rather than percentages.

This method is extended to multiple days (equity-only), where asset paths are iteratively simulated and used to compute terminal portfolio losses. These multi-step simulations serve as a forward-looking, non-analytical forecast of risk.

To better capture return asymmetry and regime switching, we implement a Gaussian Mixture Model (GMM) extension. In the multivariate case:

$$r^{(i)} \sim \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(\mu_k, \Sigma_k),$$

where K is the number of Gaussian components (typically between 2 and 4), and π_k are the mixture weights. Prices are propagated like before.

We also support a univariate GMM version, which fits a mixture to log-returns of a single asset. As with Monte Carlo, both univariate and multivariate GMMs can be extended to multiday horizons, although only for equity portfolios.

We also implement univariate GARCH-based simulations. A GARCH(1,1) model is fitted to historical log-returns, and future volatility paths are simulated under zero drift ($\mu = 0$) to avoid artificial upward trending in asset paths. Two innovation types are supported: standard normal shocks and standardized Student-t shocks for heavier tails. Prices are again obtained using the same exponential propagation. This approach serves as a multiday VaR forecast, extending naturally beyond the one-day horizon.

As a non-parametric alternative, we incorporate a suite of Historical Simulation methods with varying complexity and adaptability: standard Historical Simulation, a bootstrap-enhanced variant, Weighted Historical Simulation that emphasizes recent data, and Filtered Historical Simulation incorporating conditional volatility.

A key advantage of historical methods is that they completely bypass the need to explicitly model or estimate the portfolio's variance-covariance matrix. Instead, they implicitly preserve the dependence structure among assets by reusing actual past return vectors—i.e., the returns observed on the same historical day—when generating scenarios.

In the standard version, historical log-returns are applied directly to current prices using the same exponential propagation as in the parametric models. The bootstrap variant resamples past return vectors with replacement, which can improve robustness in small-sample settings and help reduce overfitting to specific past events.

The Weighted Historical Simulation variant improves responsiveness to recent market behavior by applying exponentially decaying weights.

This matches the approach used in hybrid VaR and ES, applying exponential weights to emphasize recent losses.

Finally, Filtered Historical Simulation combines GARCH modeling with empirical innovations. A GARCH model is fitted to each asset, and standardized residuals z_t are either directly used or resampled, then rescaled using the most recent conditional volatility $\hat{\sigma}_T$:

$$r^{(i)} = \hat{\sigma}_T \cdot z^{(i)},$$

with returns propagated to prices as before. This approach captures time-varying volatility while remaining non-parametric in distributional assumptions.

Across all methods, except for the WHS case, Expected Shortfall is computed as:

$$\mathrm{ES}_\alpha = \mathbb{E}[\ell \mid \ell \geq \mathrm{VaR}_\alpha].$$

For Weighted Historical Simulation, ES is adjusted using the same exponential decay scheme as in Hybrid VaR, ensuring that tail losses are weighted consistently with the VaR estimation.

We recognize the following limitations of our simulation framework. All methods assume static portfolio weights and do not incorporate stochastic volatility models such as Heston, or stochastic interest rate models like Vasicek or CIR. Instead, our setup is intentionally simplified and built around the Black-Scholes framework, as introduced earlier. This choice enables efficient full-valuation risk estimation while keeping the implementation practical and focused. For computational reasons, no backtesting functionality is included within the simulation methods.

10 Backtesting

Once risk models are in place, it is essential to evaluate their performance. Backtesting provides a formal framework for assessing whether the predicted VaR levels align with realized losses. Specifically, it checks the frequency and temporal structure of violations (exceptions), offering evidence on whether the model is correctly calibrated.

We implement three standard tests. The Kupiec test (unconditional coverage) evaluates if the number of observed exceptions N over T days matches the expected number under the model. If p is the failure probability, then under the null N follows a binomial distribution. The likelihood ratio test statistic is:

$$\mathrm{LR}_{\mathrm{uc}} = -2\left\{\ln\left[(1-p)^{T-N}p^N\right] - \ln\left[\left(1-\frac{N}{T}\right)^{T-N}\left(\frac{N}{T}\right)^N\right]\right\}, \quad \mathrm{LR}_{\mathrm{uc}} \sim \chi^2_1,$$

where χ^2_k denotes a chi-squared distribution with k degrees of freedom.

The Christoffersen test (conditional coverage) tests the independence of exceptions by modeling their dynamics as a first-order Markov chain. It compares the likelihoods of transition counts under the null (independence) and alternative. The statistic is:

$$\mathrm{LR_c} = -2(\ln L_0 - \ln L_1), \quad \mathrm{LR_c} \sim \chi^2_1,$$

where L_0 is the likelihood under independence and L_1 under the alternative.

The joint test combines both:

$$\mathrm{LR} = \mathrm{LR_{uc}} + \mathrm{LR_c} \sim \chi^2_2.$$

Rejecting the null in any of the tests suggests that the VaR model is either miscalibrated (too many or too few violations) or fails to capture time dependence in risk (e.g., volatility clustering).

Our back-testing framework covers every method except the simulation models.

Beyond statistical calibration, backtesting intuitively helps answer the question: "Which model is the best?" A good model is not only one that aligns with financial theory and empirical facts—such as volatility clustering, conditional heteroskedasticity, and fat tails—but also one that consistently passes validation tests. However, care must be taken to avoid overfitting to a particular dataset or test. For this reason, stress testing is also commonly used in practice as a complementary tool, although it is not automatically supported in the current version of the package.

11 Results and Applications

Virtually all applications and results of the `pyvar` package can be found in the `examples` folder of the GitHub repository. A series of Jupyter notebooks illustrate typical use cases, all following a consistent setup: the creation of a simple portfolio followed by the use of the package's functions to calculate various risk metrics and visualize them effectively. Each notebook includes additional insights and brief theoretical refreshers to support understanding and contextualize the results.

In addition to the notebooks, we report here selected example outputs and backtesting results. For univariate models, we analyze an investment of 100,000 USD in the S&P 500 from 4 January 2000 to 31 December 2024. Using the parametric normal model, the estimated 1-day monetary VaR at the 99% confidence level is 2,846.04 USD. However, the model fails backtesting on all fronts: *Total Violations*: 121, *Violation Rate*: 1.92%, and it fails both the joint and independence tests. A plot of the backtesting violations is shown in Figure 1. In contrast, the same experiment using a GJR-GARCH(1,1) model with skew-t innovations produces a more realistic, conditional VaR estimate, passing all backtesting tests with *Total Violations*: 63 and *Violation Rate*: 1.00%. The corresponding backtesting plot, shown in Figure 2, focuses on a high-volatility subset of the full sample period.

We also report results for a globally diversified ETF and stock portfolio constructed from long and short positions in assets including VT, SPY, VGK, EWJ, VWO, AAPL, JNJ, PG, UNH, NVS, and a short position in XLI. The portfolio is held over the period from 3 January 2023 to 31 December 2024, and the total equity value of 114,633.77 USD refers to the final value of the portfolio's equity component on the last day.

Using both the single-factor model (with the S&P 500 as benchmark, APARCH(1,1) volatility) and the Fama–French three-factor model, the backtest yields *Total Violations*: 5 and *Violation Rate*: 1.00%, with all tests passed. Figure 3 shows the estimated portfolio volatility under both specifications.

For time-varying correlation models, we backtest four approaches: moving average with a 60-day window yields *Total Violations*: 5 and *Violation Rate*: 1.13%; EWMA with $\lambda = 0.94$ gives *Total Violations*: 6 and *Violation Rate*: 1.20%; rolling PCA with three components and Ledoit–Wolf shrinkage (both with a 20-day window) each yield *Total Violations*: 5 and *Violation Rate*: 1.04%. All models pass the joint, unconditional, and independence tests. Empirical innovations are used throughout. Figure 4 displays the portfolio volatility from the four correlation models.

For simulation-based methods, we augment the previous portfolio by including three European-style options: a long call on AAPL (3,000 contracts), a long put on XLI (5,000 contracts), and a long call on VT (2,500 contracts), all priced using the Black–Scholes formula at time 0 (option values include the standard contract multiplier of 100). The respective values are 15,042,544.75 USD , 18,396.70 USD, and 4,235,811.88 USD, resulting in a total option value of 19,296,753.33 USD and a total portfolio value of 19,411,387.10 USD. The following 1-day VaR estimates at the 99% confidence level are obtained: Monte Carlo simulation yields 4,407,931.13 USD; bootstrapped historical gives 4,556,184.04 USD; Filtered Historical Simulation (FHS) produces 7,158,684.58 USD; Weighted Historical Simulation (WHS) gives 6,508,074.85 USD; and GMM-based Monte Carlo results in 4,618,421.87 USD. Figure 5 shows the KDE plots of simulated profit and loss (P&L) distributions and the corresponding VaR estimates for each simulation method.

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- Jeff Fleming, Chris Kirby, and Barbara Ostdiek, *The Economic Value of Volatility Timing Using Realized Volatility*, Journal of Financial Economics, Vol. 67, No. 3, 2003, pp. 473–509.

Appendix

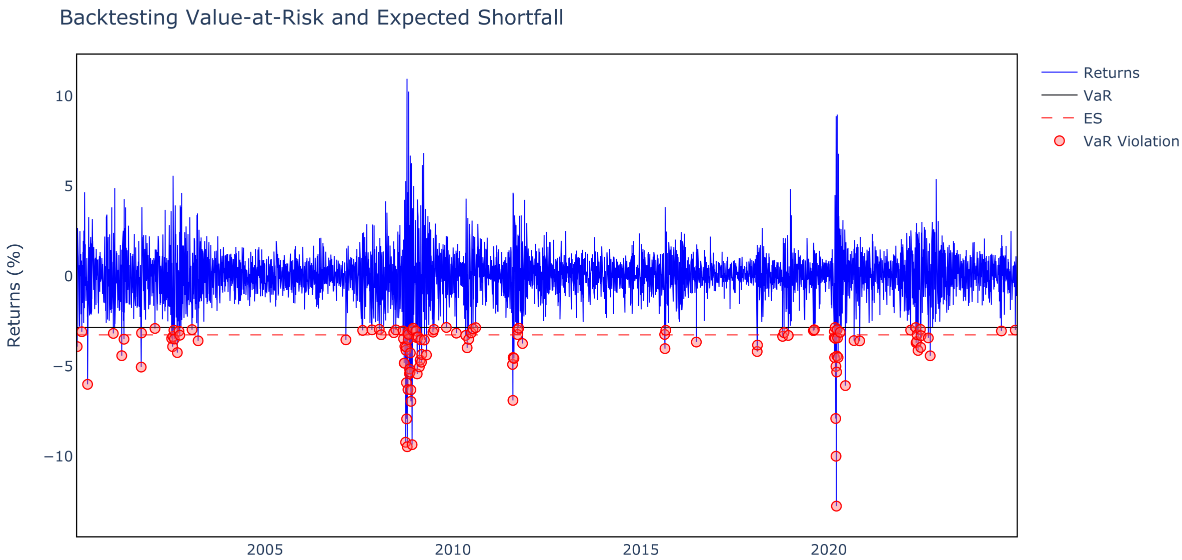

Figure 1: Backtesting plot for the normal model applied to the S&P 500 (2000–2025).

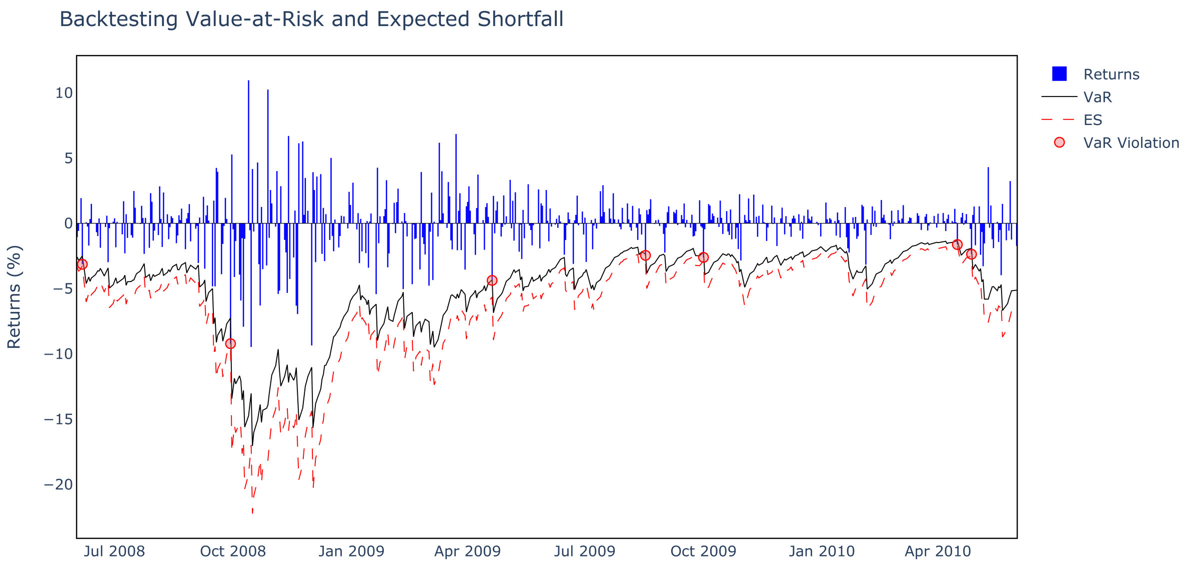

Figure 2: Backtesting plot for the GJR-GARCH(1,1) model on a high-volatility subset.

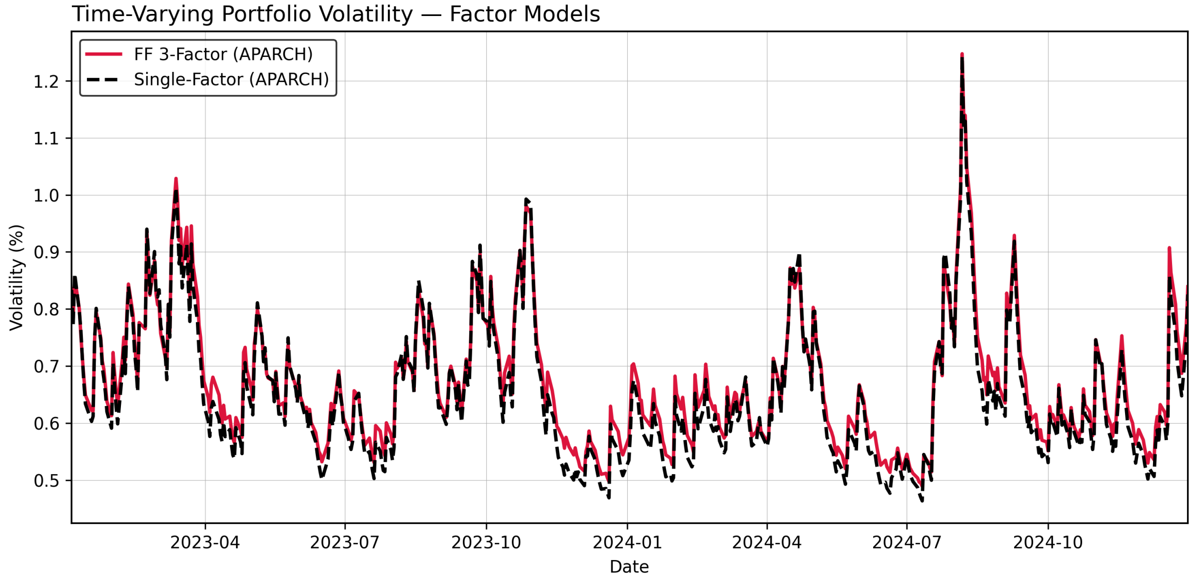

Figure 3: Estimated portfolio volatility under the single-factor and Fama–French three-factor models.

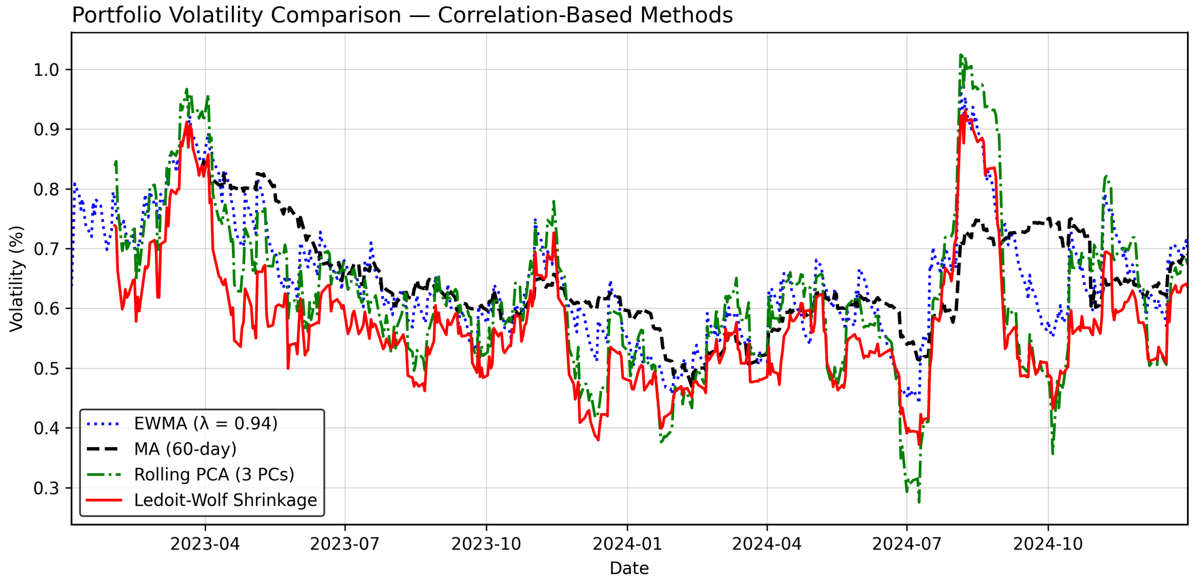

Figure 4: Portfolio volatility estimated using MA, EWMA, PCA, and Ledoit–Wolf correlation models.

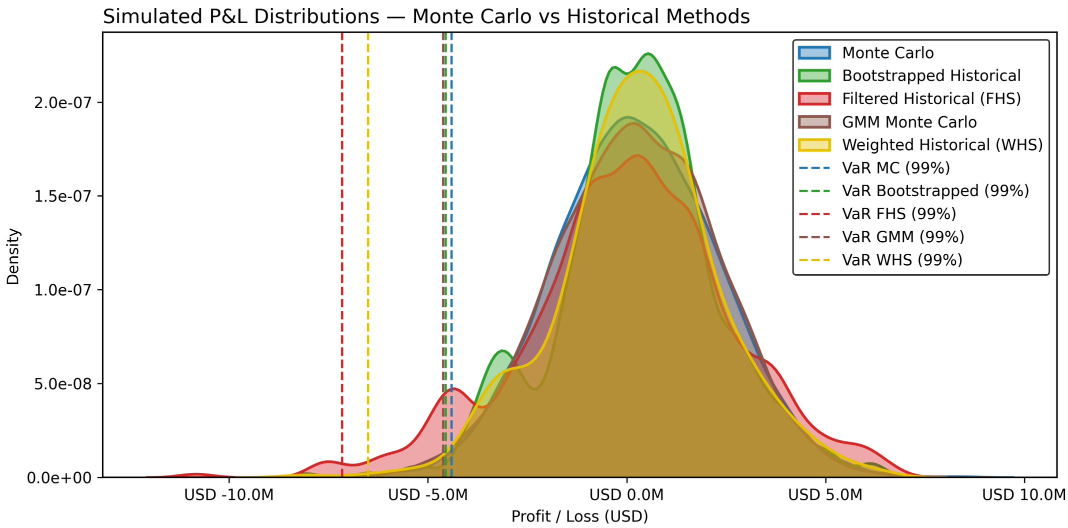

Figure 5: KDEs of simulated profit and loss with 1-day VaR for each simulation method.